
Digital Signal Processing

Lecture 2

Signal Representation and Time Domain Analysis

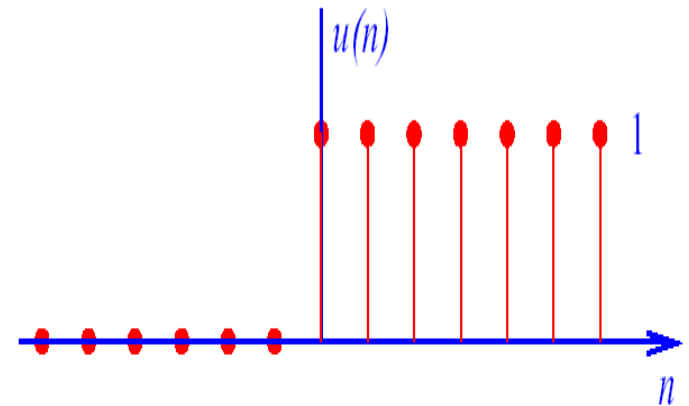
Dr. Tahir Zaidi

Basic Types of Digital Signals

The unit-step and unit-impulse functions

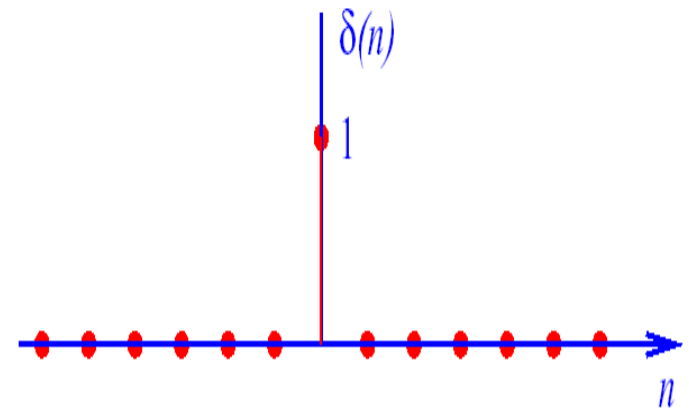
Unit-step function:

$$u(n) = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases}$$



Unit-impulse function:

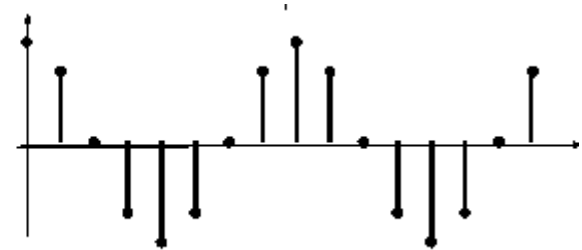
$$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$



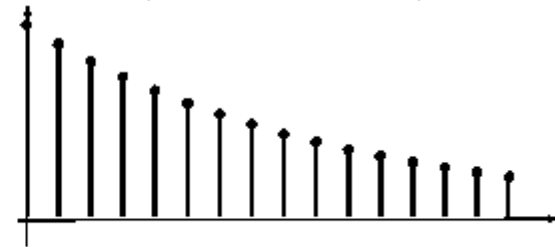
$$\delta(n) = u(n) - u(n - 1)$$

Basic Types of Digital Signals

sinusoidal sequence $\cos[\omega_0 n]$



exponential sequence a^n



Sine and Exp Using Matlab

% sine generation: $A \cdot \sin(\omega n + \theta)$

% exponential generation: A^n

$n = 0: 1: 50;$

% amplitude

$A = 0.87;$

% phase

$\theta = 0.4;$

% frequency

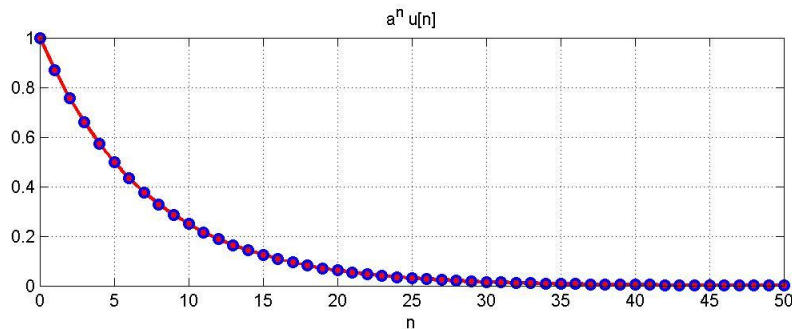
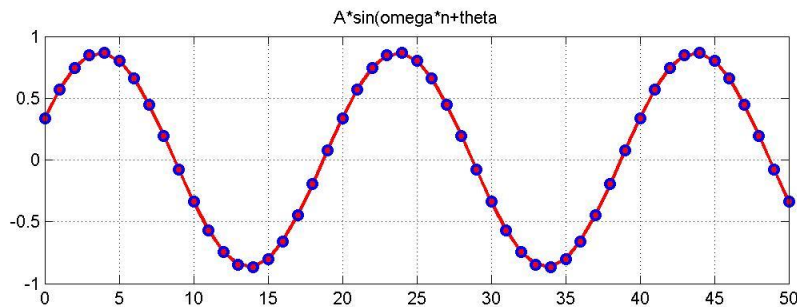
$\omega = 2\pi / 20;$

% sin generation

$xn1 = A \cdot \sin(\omega n + \theta);$

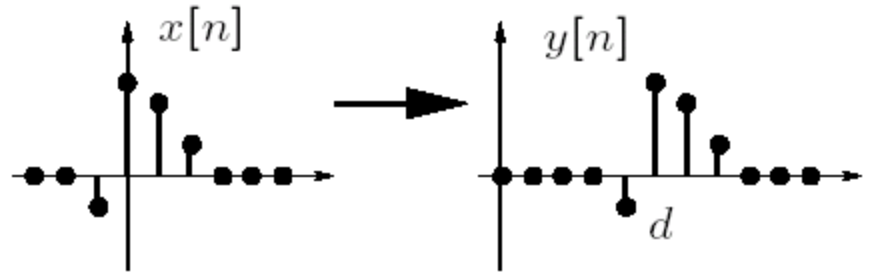
% exp generation

$xn2 = A.^n;$

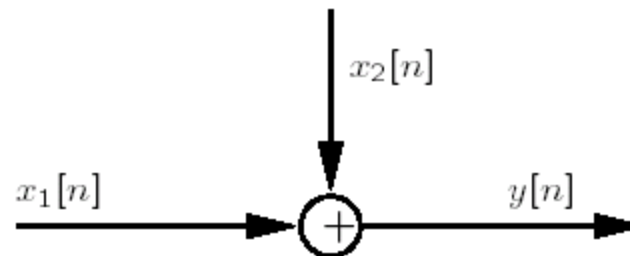


Basic Operations

- ideal delay: $y[n] = x[n - d]$

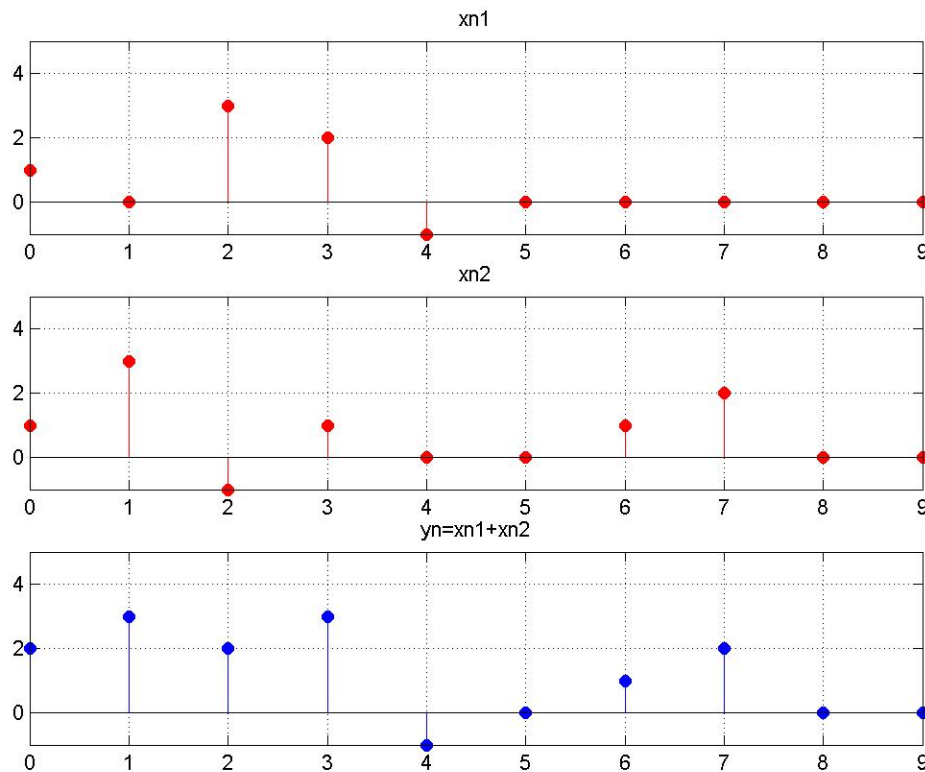


- sum, difference:
 $y[n] = x_1[n] \pm x_2[n]$



- multiplication: $y[n] = x_1[n]x_2[n]$

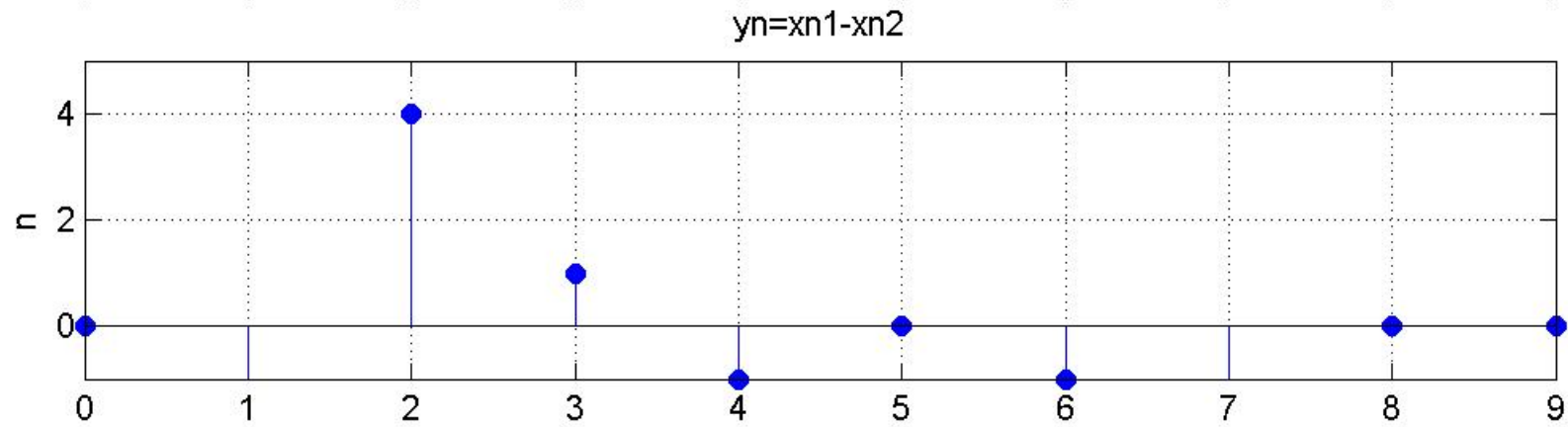
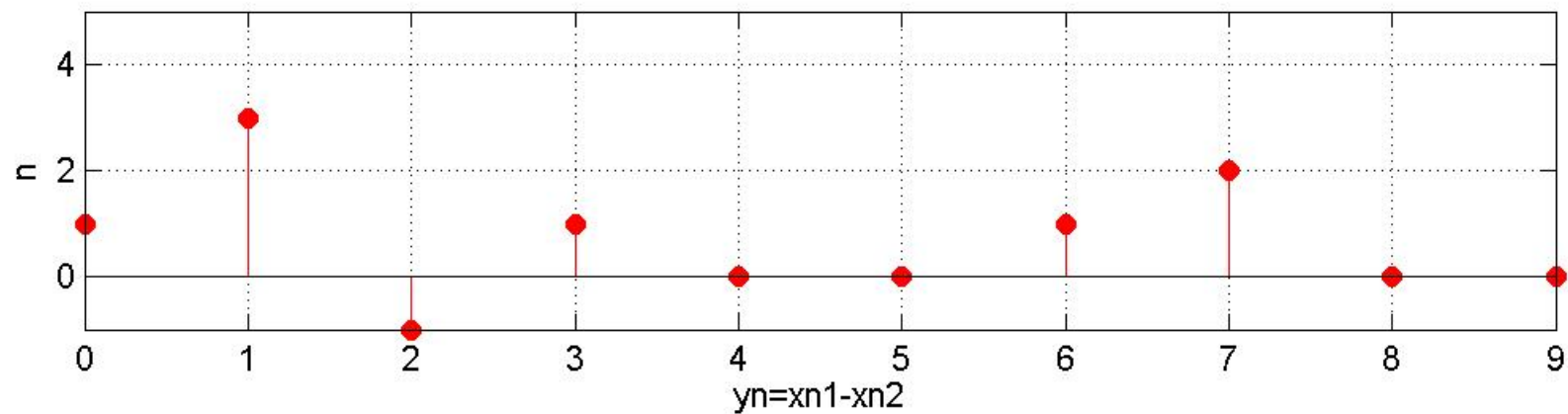
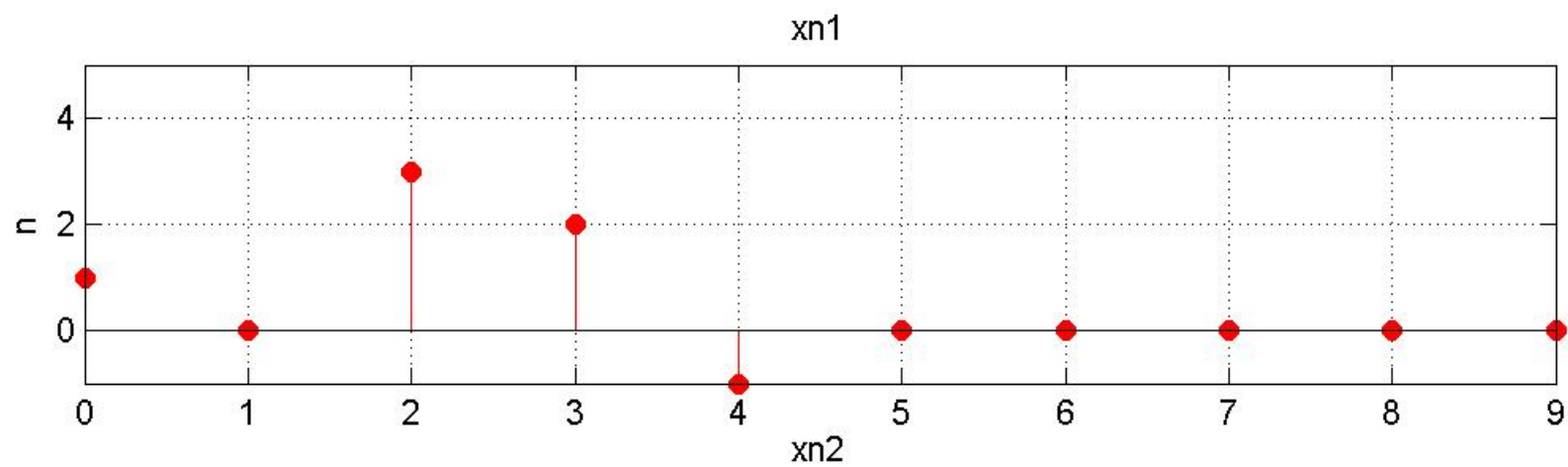
Operations in Matlab



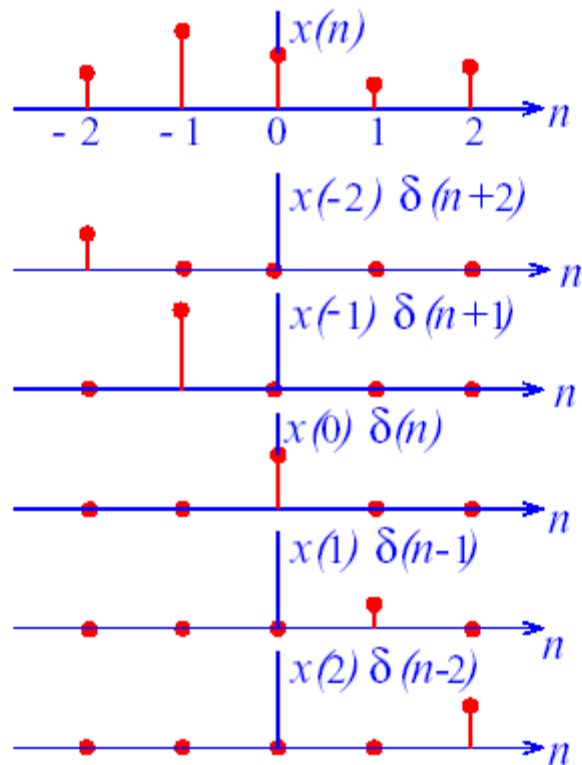
$xn1 = [1 \ 0 \ 3 \ 2 \ -1 \ 0 \ 0 \ 0 \ 0 \ 0];$

$xn2 = [1 \ 3 \ -1 \ 1 \ 0 \ 0 \ 1 \ 2 \ 0 \ 0];$

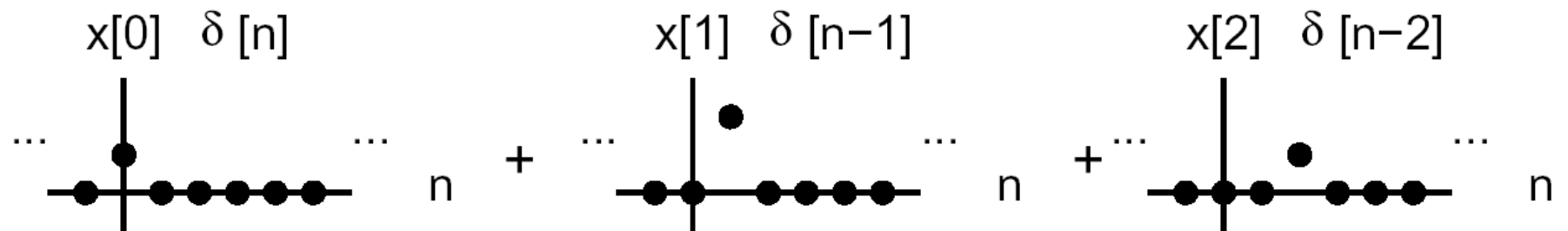
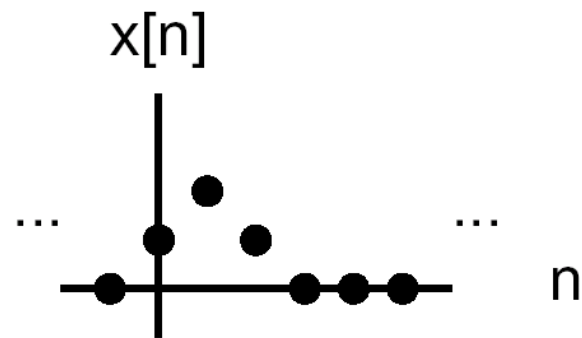
$yn = xn1 + xn2;$



$x[n]$ via impulse functions



$$x(n) = \sum_{k=-\infty}^{\infty} x(k) \delta(n - k)$$



Time Domain Analysis

Linear Time-Invariant Systems

Definition of a system:

$$y(n) = T\{x(n)\}$$

where $T\{\cdot\}$ is an operator that maps an input sequence $x(n)$ into an output sequence $y(n)$.

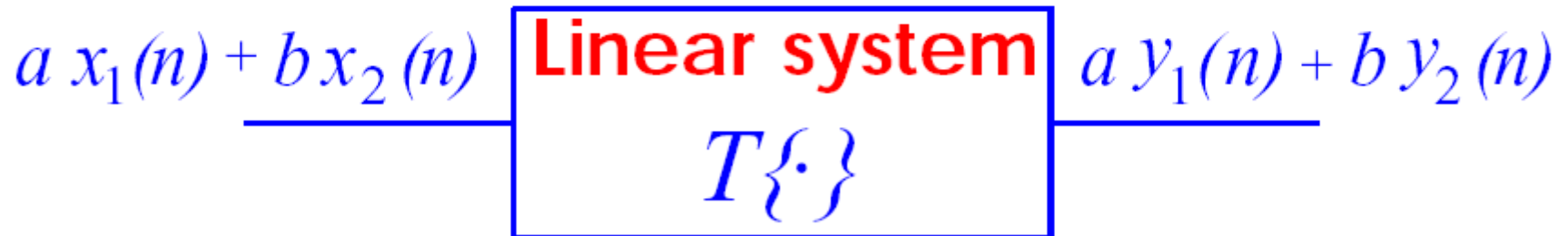
Linear system: A system (or processor) is **linear** if it obeys the **principle of superposition**.

Principle of superposition: If the input of a system contains the sum of multiple signals then the output of this system is the **sum of the system responses** to each separate signal.

Linear

A system is linear if and only if:

$$\begin{aligned} T\{ax_1(n) + bx_2(n)\} &= aT\{x_1(n)\} + bT\{x_2(n)\} \\ &= ay_1(n) + by_2(n) \end{aligned}$$



Example: Let $y(n) = x^2(n)$ (i.e., $T\{\cdot\} = (\cdot)^2$). Then,

$$\begin{aligned} T\{x_1(n) + x_2(n)\} &= x_1^2(n) + x_2^2(n) + 2x_1(n)x_2(n) \\ &\neq x_1^2(n) + x_2^2(n) \end{aligned}$$

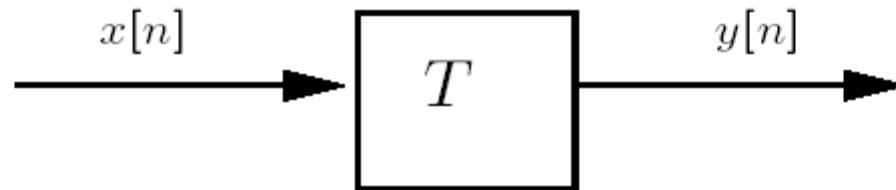
Hence, this system is **nonlinear**!

A time-invariant system has properties unvarying with time, i.e.:

$$\text{if } y(n) = T\{x(n)\} \implies y(n - k) = T\{x(n - k)\}$$

Linear Time-Invariant (LTI) system is a system that is both linear and time-invariant (sometimes referred to as a **Linear Shift-Invariant (LSI)** system)

Linear Time-Invariant System



Ex:

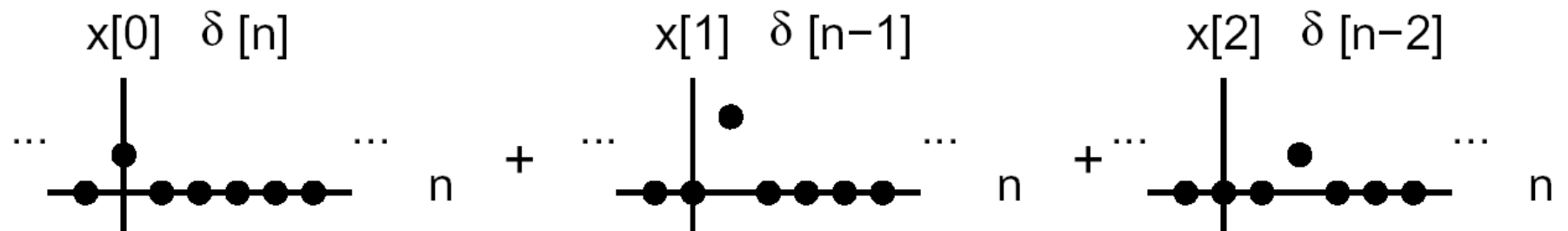
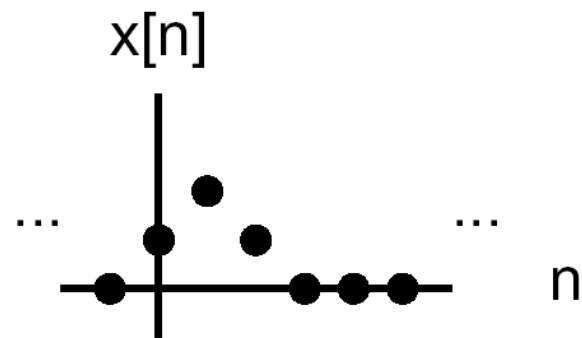
- ideal delay system: $y[n] = x[n - d]$
- Moving average system: $y[n] = \frac{1}{M + N + 1} \sum_{k=-M}^N x[n - k]$

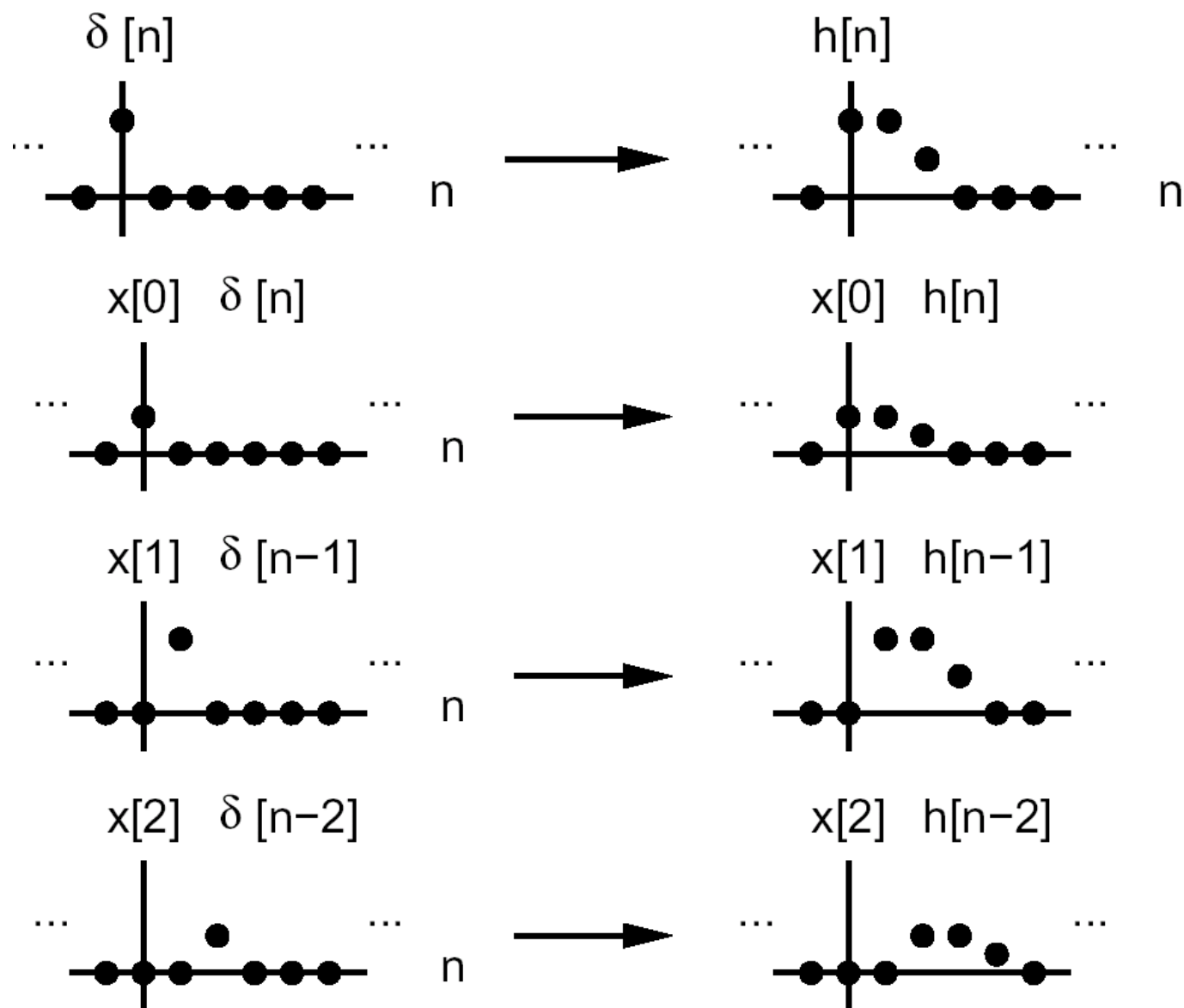
Linear Time-Invariant System

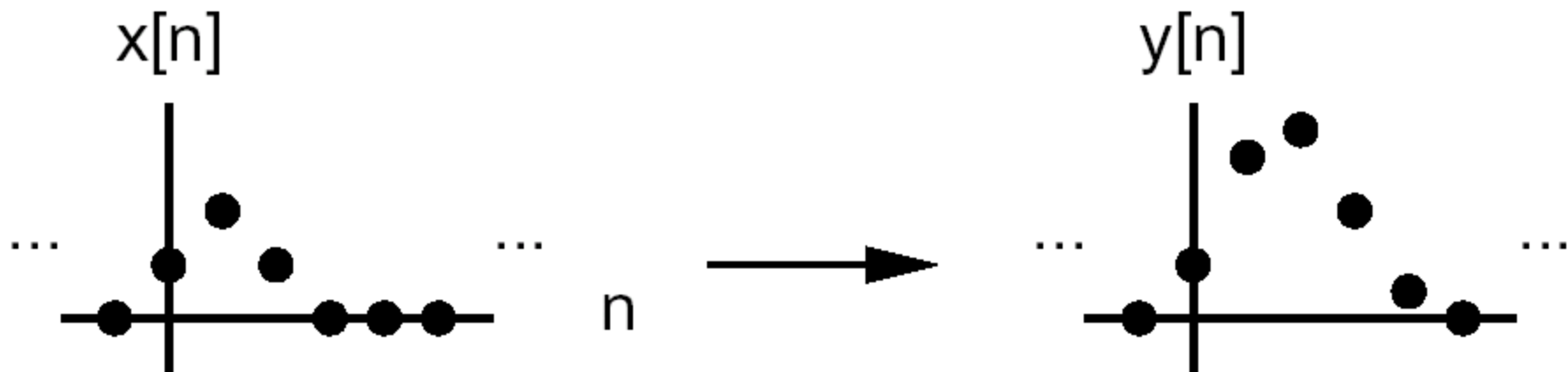
Let $h(n)$ be the response of the system to $\delta(n)$.

Due to the time-invariance property, the response to $\delta(n - k)$ is simply $h(n - k) \implies$

$$\begin{aligned} y(n) &= T\{x(n)\} \\ &= T\left\{\sum_{k=-\infty}^{\infty} x(k)\delta(n - k)\right\} \\ &= \sum_{k=-\infty}^{\infty} x(k)T\{\delta(n - k)\} \\ &= \sum_{k=-\infty}^{\infty} x(k)h(n - k) = \{x(n)\} * \{h(n)\} \quad \text{convolution sum} \end{aligned}$$

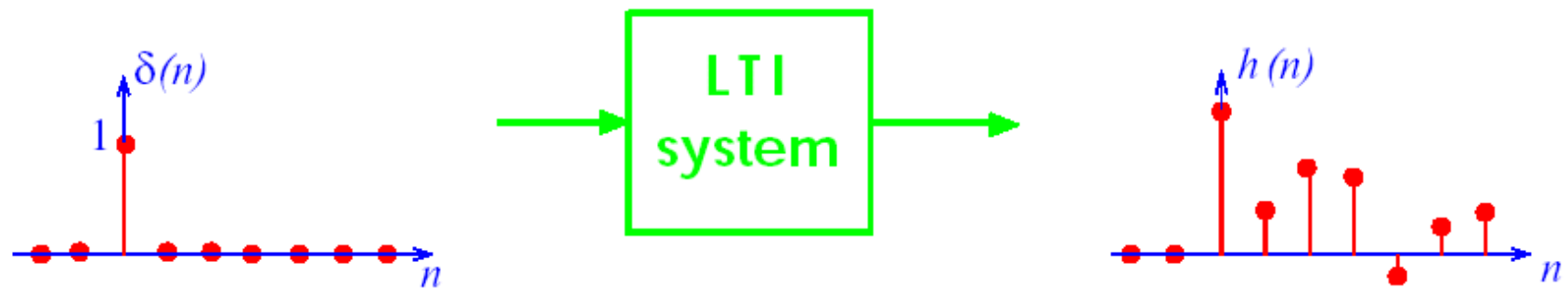






LTI System

The sequence $\{h(n)\}$ is commonly referred to as **impulse response** of the LTI system



An important property of convolution:

$$\begin{aligned}\{x(n)\} * \{h(n)\} &= \sum_{k=-\infty}^{\infty} x(k)h(n-k) = \sum_{k=-\infty}^{\infty} h(k)x(n-k) \\ &= \{h(n)\} * \{x(n)\} \quad \Rightarrow\end{aligned}$$

the **order** in which two sequences are convolved is **unimportant!**

Two ways

- As the representation of the output as a sum of delayed and scaled impulse responses.
- As a computational formula for computing $y[n]$ (“y at time n”) from the entire sequences x and h .
 - Form $x[k]h[n-k]$ for $-\infty < k < +\infty$ for a fixed n
 - Sum over all k to produce $y[n]$

Convolution in the time domain:

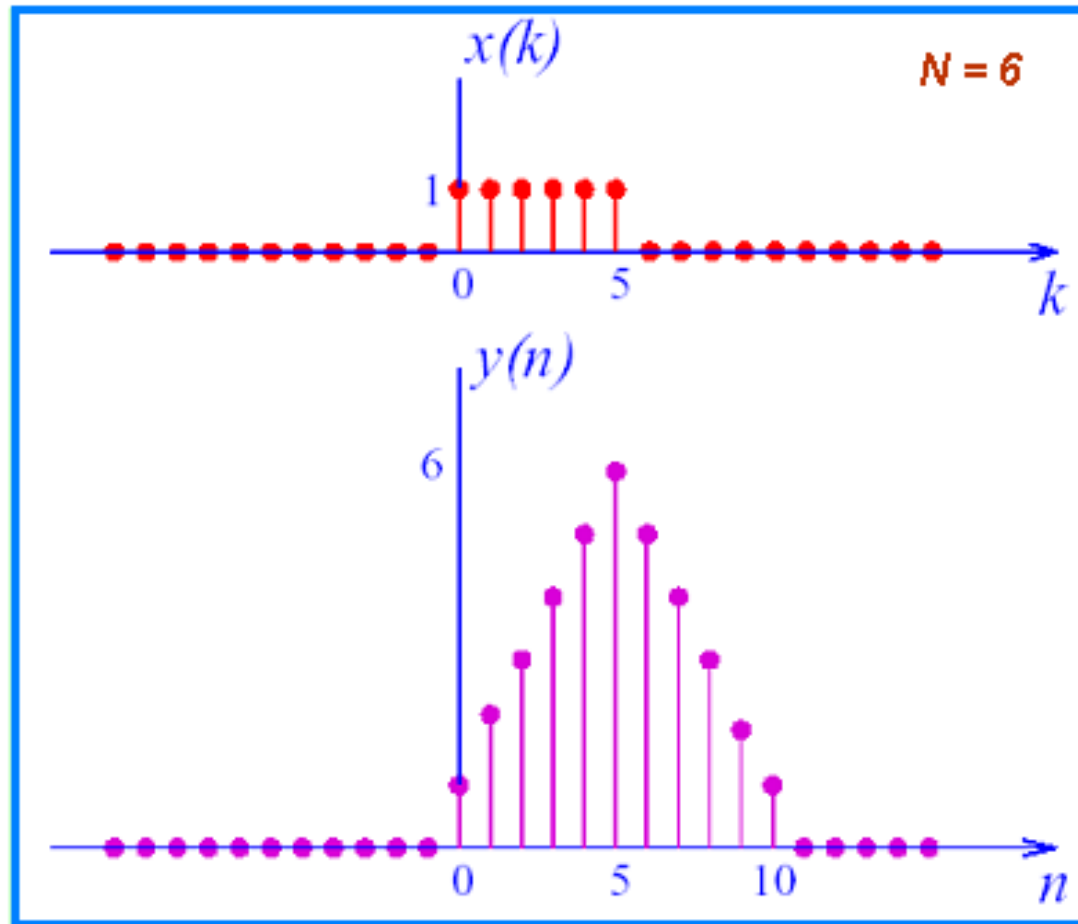
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

n			0	1	2	3	4	5	6	7	8	
x[n]			1	-2	3	-1	-1	0	0	0	0	
h[n]	-1	1	2									
y[0]			2									
x[n]			1	-2	3	-1	-1	0	0	0	0	
h[n]		-1	1	2								
y[1]				-3								
x[n]			1	-2	3	-1	-1	0	0	0	0	
h[n]			-1	1	2							
y[2]					3							

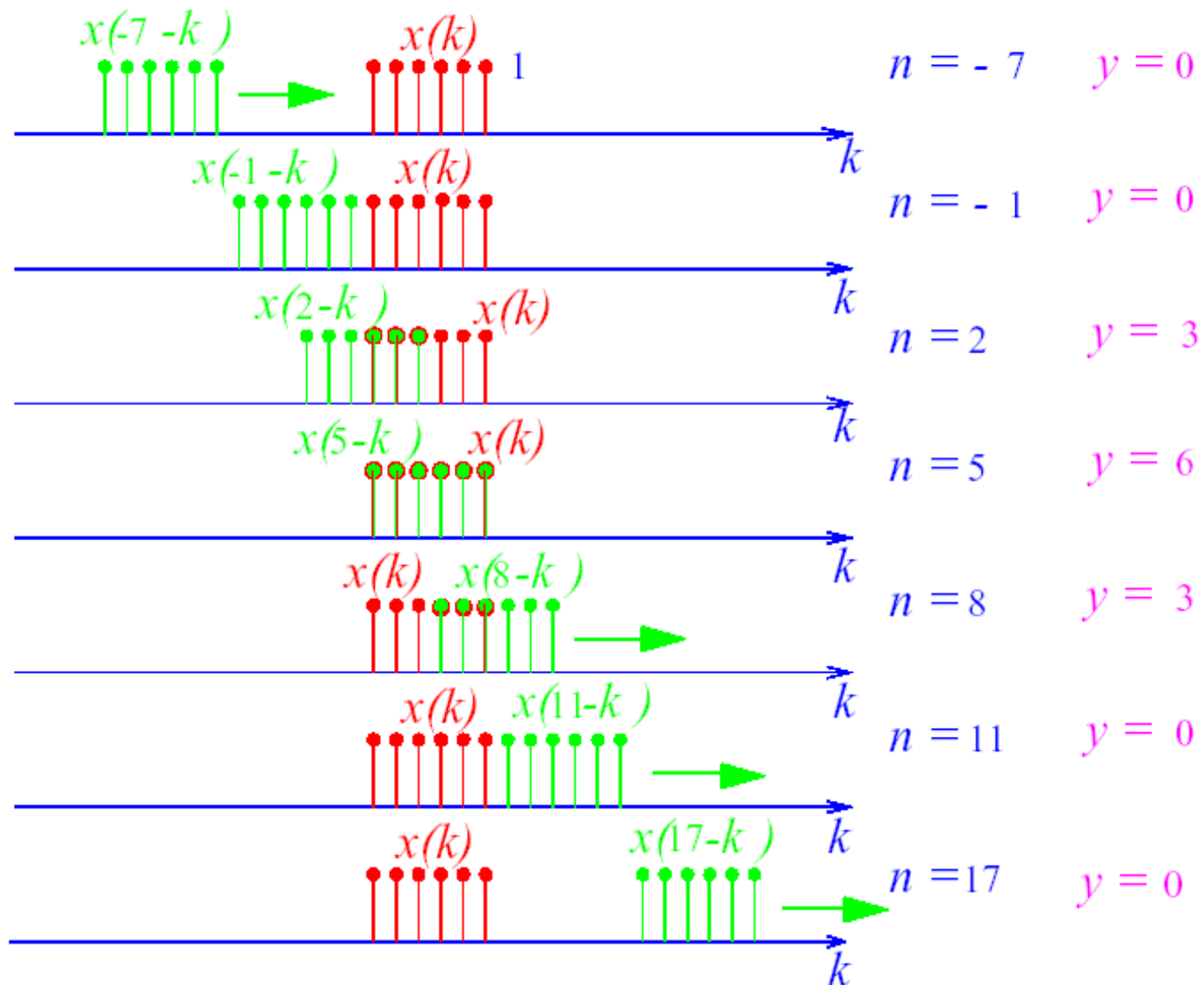
$$y[n] = 2 \ -3 \ 3 \ 3 \ -6 \ 0 \ 1 \ 0 \ 0$$

Example–Convolution of Two Rectangles

$$y(n) = \{x(n)\} * \{x(n)\}$$



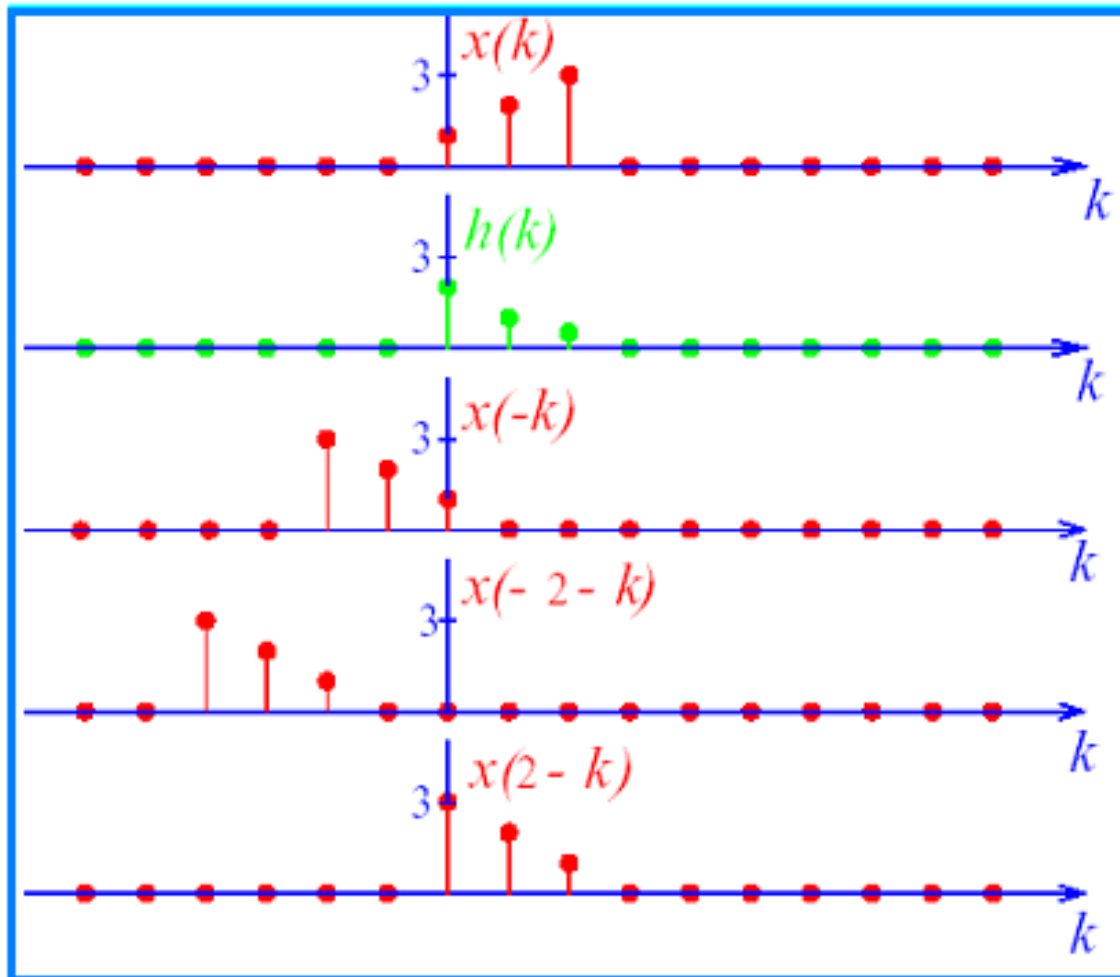
Example..(Continued)

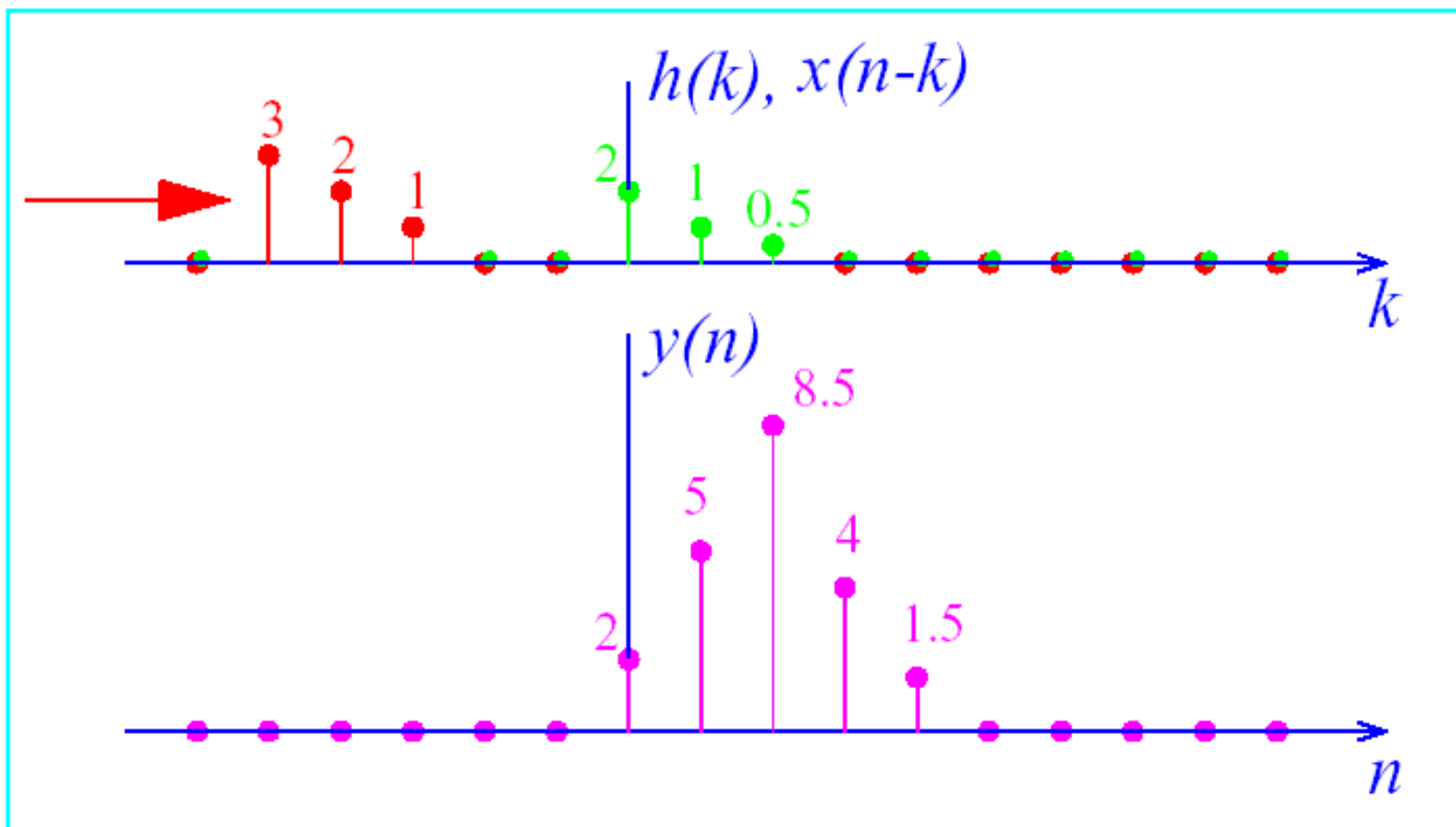


Example–Convolution Of Two Sequences

$$\{x(n)\} = \{\dots, 0, 1, 2, 3, 0, \dots\}$$

$$\{h(n)\} = \{\dots, 0, 2, 1, 0.5, 0, \dots\}$$





Stability

Result: LTI systems are **stable** if and only if

$$\sum_{k=-\infty}^{\infty} |h(k)| < \infty$$

Proof of "if": Let the input $x(n)$ be bounded so that $|x(n)| < L_x$,
 $\forall n \in [-\infty, \infty]$. Then

$$\begin{aligned} |y(n)| &= \left| \sum_{k=-\infty}^{\infty} h(k)x(n-k) \right| \\ &\leq \sum_{k=-\infty}^{\infty} |h(k)||x(n-k)| \\ &\leq L_x \sum_{k=-\infty}^{\infty} |h(k)| \implies |y(n)| < \infty \text{ if } \sum_{k=-\infty}^{\infty} |h(k)| < \infty \end{aligned}$$

Causality

Definition: A causal system is one for which the output $y(n)$ depends on the inputs $\{\dots, x(n-2), x(n-1), x(n)\}$ only.

Result: An LTI system is causal if and only if its impulse response $h(n) = 0$ for $n < 0$.

Proof of "if": From the definition of a causal system,

$$\begin{aligned} y(n) &= \sum_{k=-\infty}^{\infty} h(k)x(n-k) \\ &= \sum_{k=0}^{\infty} h(k)x(n-k) \end{aligned}$$

Obviously, this equation is valid if $\sum_{k=-\infty}^{-1} h(k)x(n-k) = 0$ i.e., if $h(n) = 0$ for $n < 0$.

Causality & Stability– Example

Example: An LTI system with

$$h(n) = a^n u(n) = \begin{cases} a^n, & n \geq 0 \\ 0, & n < 0; \end{cases}$$

- since $h(n) = 0$ for $n < 0$, the system is **causal**
- To decide on stability, we must compute the sum

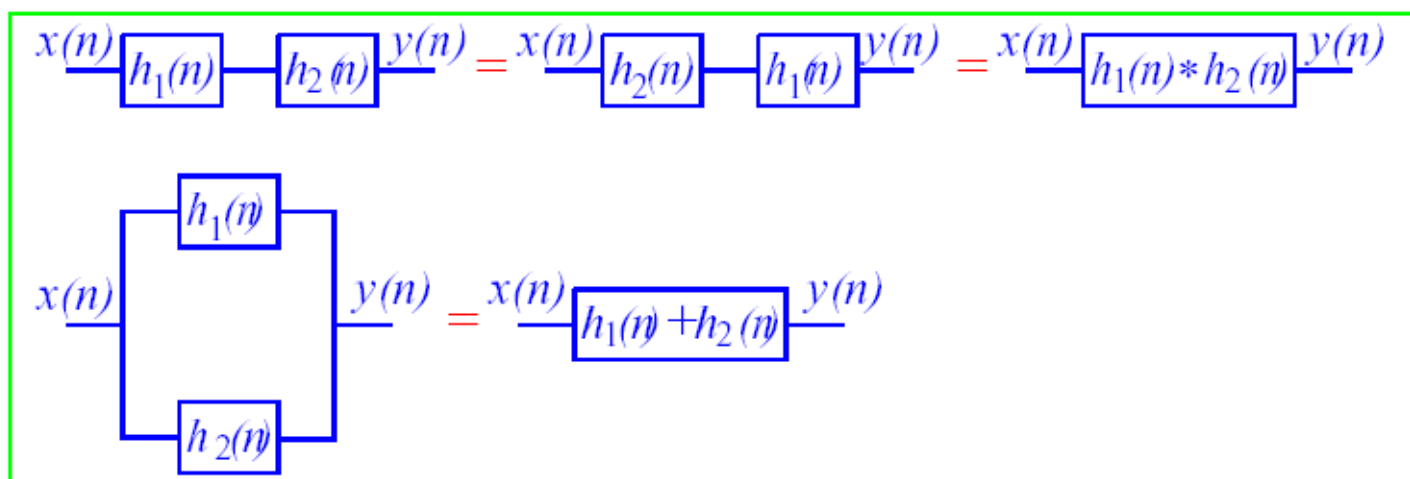
$$S = \sum_{k=-\infty}^{\infty} |h(k)| = \sum_{k=0}^{\infty} |a|^k = \begin{cases} \frac{1}{1-|a|}, & |a| < 1 \\ \infty, & |a| \geq 1; \end{cases} \quad \Rightarrow$$

the system is stable only for $|a| < 1$

Other properties of convolution:

$$\begin{aligned} & \{x(n)\} * [\{h_1(n)\} * \{h_2(n)\}] \\ = & [\{x(n)\} * \{h_1(n)\}] * \{h_2(n)\} \end{aligned} \quad \text{associativity}$$

$$\begin{aligned} & \{x(n)\} * [\{h_1(n)\} + \{h_2(n)\}] \\ = & \{x(n)\} * \{h_1(n)\} + \{x(n)\} * \{h_2(n)\} \end{aligned} \quad \text{distributivity}$$



Difference Equation

- For all computationally realizable LTI systems, the input and output satisfy a difference equation of the form

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

- This leads to the recurrence formula

$$y[n] = -\sum_{k=1}^N \left(\frac{a_k}{a_0} \right) y[n-k] + \sum_{k=0}^M \left(\frac{b_k}{a_0} \right) x[n-k]$$

which can be used to compute the “present” output from the present and M past values of the input and N past values of the output